

Answer Key: Computers and Their Development Worksheet

Teacher's copy — do not distribute to students

Section	Tasks	Marks	Notes
Part A — Comprehension	A1, A2, A3	20 marks	1 mark per correct answer
Part B — Vocabulary	B1, B2	20 marks	1 mark per correct answer
TOTAL		40 marks	

PART A — COMPREHENSION ANSWERS

Task A1 — True / False / Not Given [7 marks]

Accept only TRUE / FALSE / NOT GIVEN (or T / F / NG). No partial marks.

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|----|--------------|---|
| 1. | FALSE | The article states they were built for 'military and scientific problems' — not entertainment or commerce. |
| 2. | TRUE | Part One: 'These early computers stored and processed data using binary code.' |
| 3. | FALSE | The article states the transistor was invented in 1947 — two years after the war ended in 1945. |
| 4. | TRUE | Part Two: 'engineers discovered how to place multiple transistors on a single piece of semiconductor material.' |
| 5. | TRUE | Part Two: 'Moore's Law: the number of transistors ... was doubling approximately every two years.' |
| 6. | FALSE | The article states: 'It was not the first mobile phone, nor the first smartphone.' |
| 7. | TRUE | Part Five: AI tools 'reached one hundred million users in just two months' versus twenty years for the internet to reach two billion. |

Task A2 — Note Completion [9 marks — one word only]

Accept only the exact word from the article. Do not accept synonyms unless noted.

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| 8. | 1945 | Timeline: 'ENIAC — Built ... 1945.' |
| 9. | tonnes | Part One: 'It weighed around 27 tonnes.' |
| 10. | binary | Part One: 'computers stored and processed data using binary code.' |

11.	vacuum	Part Two: 'It replaced the much larger, hotter, and less reliable vacuum tube.'
12.	semiconductor	Part Two: 'on a single piece of semiconductor material — typically silicon.'
13.	years	Part Two: 'doubling approximately every two years.'
14.	1991	Part Four: 'In 1991, a British scientist named Tim Berners-Lee made the World Wide Web available.'
15.	cloud	Part Four: 'what we now call the cloud.'
16.	data	Part Five: 'these systems ... learn from examples, identifying patterns in data.'

Task A3 — Short Answer Questions [4 marks — 1 mark each]

Award 1 mark for any answer clearly conveying the key idea in the student's own words. Do not penalise grammar errors if meaning is clear.

17.	Because it replaced the vacuum tube and made smaller, cheaper, and more reliable computers possible — without it, there would be no microprocessor, personal computer, smartphone, or internet.	Key idea: transistor enabled everything that followed — accept any 2 specific consequences listed.
18.	Personal computers were small and cheap enough for individuals to own, came with software ordinary people could use, and did not require specialist knowledge to operate.	Key idea: affordable / individual ownership / no specialist knowledge needed.
19.	The internet transformed computers from standalone machines into connected devices — part of a global network where people could share information, communicate, and access data stored remotely.	Key idea: connected / networked / global access to information. Accept mention of the 'cloud'.
20.	Traditional algorithms follow a fixed set of instructions written by a programmer; AI systems learn from data, identifying patterns without being explicitly programmed.	Key idea: AI learns from data / patterns rather than following fixed instructions.

PART B — VOCABULARY ANSWERS

Task B1 — Vocabulary Matching [8 marks]

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|----|---------------------------|--|
| 1. | c — transistor | <i>A tiny electronic switch that controls electrical signals in a circuit.</i> |
| 2. | d — binary code | <i>A system that represents all information as a series of 0s and 1s.</i> |
| 3. | e — microprocessor | <i>A complete computer processor contained on a single silicon chip.</i> |
| 4. | f — mainframe | <i>A large, expensive computer used by organisations before personal computers.</i> |
| 5. | b — algorithm | <i>A set of step-by-step instructions that tells a computer how to solve a problem.</i> |
| 6. | a — hardware | <i>All the physical components of a computer — chips, circuits, screens, keyboards.</i> |
| 7. | h — software | <i>The programs and instructions that tell the hardware what to do.</i> |
| 8. | g — semiconductor | <i>A material such as silicon that can conduct electricity under certain conditions.</i> |

Task B2 — Gap-Fill [12 marks — 1 mark per gap]

Award 1 mark per blank filled correctly. Sentences 11, 12, 14, 15, and 17 require 2–3 answers — mark each blank independently.

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|-----|---|--|
| 9. | binary code | <i>1 blank.</i> |
| 10. | transistor | <i>1 blank.</i> |
| 11. | integrated circuit / semiconductor | <i>2 blanks — 1 mark each.</i> |
| 12. | microprocessor / processor | <i>2 blanks — 1 mark each.</i> |
| 13. | mainframe | <i>1 blank.</i> |
| 14. | algorithm / programming language | <i>2 blanks — 1 mark each.</i> |
| 15. | operating system / hardware / software | <i>3 blanks — 1 mark each. Accept any order for hardware and software.</i> |
| 16. | artificial intelligence | <i>1 blank.</i> |
| 17. | innovation | <i>1 blank.</i> |

MARKING SUMMARY

A1 (T/F/NG): 7 marks · A2 (Note completion): 9 marks · A3 (Short answer): 4 marks · B1 (Matching): 8 marks · B2 (Gap-fill): 12 marks · TOTAL: 40 marks

